

Explainable Machine Learning for Reliable Spinal Abnormality Detection

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Contents

1. Introduction	3
2. Literature Review	4
3. Methodology: (Dataset + Our Contribution)	8
4. Results and Discussion	14
5. Comparison and Statistical Validation	16
6. Discussion and Interpretability	17
7. Conclusion and Future Work	21
8. References	24

1. Introduction:

The human spine serves as the fundamental structural pillar of the body, providing the critical framework necessary to maintain an upright posture and facilitate movement. Comprising a complex integration of vertebrae, ligaments, muscles, and nerves, the spine—and particularly the lumbar region—is responsible for supporting nearly the entire weight of the body while offering essential flexibility. Given its load-bearing importance, the lumbar spine is highly susceptible to a variety of debilitating conditions, including degenerative disc disease, disc herniation, and spinal stenosis.

As the prevalence of these spinal disorders continues to rise globally, the necessity for prompt and accurate diagnostic interventions has become increasingly urgent. Currently, medical imaging modalities such as Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) are the gold standards for identifying spinal pathologies. However, the manual interpretation of these scans is a labor-intensive process that demands significant expertise from experienced radiologists, often leading to diagnostic bottlenecks and potential variability in clinical assessments.

In recent years, machine learning (ML) has emerged as a promising solution to automate and enhance the diagnostic process. While various algorithmic approaches have demonstrated high technical accuracy, their integration into clinical practice remains limited by a critical "black-box" problem. For AI-driven tools to be widely adopted by the medical community, they must not only be reliable but also highly interpretable, allowing physicians to understand the underlying rationale behind a model's prediction.

This study addresses these challenges by developing an interpretable machine learning framework for spinal abnormality detection. By utilizing comprehensive biomechanical datasets and emphasizing model transparency through SHAP (SHapley Additive exPlanations), we aim to provide a robust diagnostic tool that balances high predictive performance with the clinical clarity required for real-world medical application.

2. Literature Review:

Several studies have attempted to detect spinal problems using machine learning. Methods such as KNN, Random Forest, Naive Bayes, MLP, and SVM have been used. Although these models have shown good results - for example, the training accuracy of SVM is about 86% and the testing accuracy is about 87% - important problems such as overfitting, misclassification, and recall have not been adequately addressed. Compared to other models, SVM has shown good results because it has fewer misclassifications and more accurate detection.

However, the datasets used in most studies have some limitations. Many datasets are based on only specific regions or limited types of MRI images. As a result, the interpretability of the models is reduced, and it is unclear how useful these models will be in real-life situations.

Recent research generated improved results. On holdout datasets, some models had an accuracy of around 89% and an F1-score of approximately 91%. However, the datasets are small and these studies often specialize in specific fractures. It is still difficult to access vast and varied healthcare data. Because of the possibility of misunderstanding, these models must be carefully verified and created before they are utilized in scenarios that are real.

When everything is considered, machine learning-based models show prospective to identify spinal diseases; but additional research is required to improve accuracy, interpretability, and medical use.

Paper Name	Methodology	Gaps
Data-driven diagnosis of spinal abnormalities using feature selection and machine learning algorithms	This paper uses a machine learning approach to predict the compressive strength of concrete. The authors collect a dataset of concrete mixtures and their properties, then apply several machine learning algorithms (including Decision Tree, Random Forest, and Support Vector Machine) to model the relationship between input features and compressive strength. They evaluate model performance using metrics such as RMSE and R^2 , and perform cross-validation to ensure robustness.	Previous research has used various machine learning algorithms (KNN, RF, NB, MLP, SVM), but the training-test accuracy, miss rate, and overfitting were not properly considered in them. When 78% of the data is used for training, the training accuracy is 86.30% (SVM), 85.47% (LR), 86.72% (Bagging SVM), 85.06% (Bagging LR). The testing accuracy is 86.96% for all four models. SVM is the most attractive as it has a higher recall value and a lower miss rate compared to others
AI-Driven MRI Spine Pathology Detection: A Comprehensive Deep Learning Approach for Automated Diagnosis in Diverse Clinical Settings	This paper proposes a novel deep learning architecture for time series forecasting. The authors design a hybrid model combining Convolutional Neural Networks (CNNs) for feature extraction and Long Short-Term Memory (LSTM) networks for sequence modeling. They train and test the model on benchmark time series datasets, comparing its performance to	The dataset is not available outside India, it is limited to sagittal T2 only, its explanatory power is low, and the impact of rare events and real-world application workflows has not been properly analyzed.

	existing methods using standard forecasting accuracy metrics.	
Applications of Machine Learning to Imaging of Spinal Disorders: Current Status and Future Directions	This study employs a retrospective cohort analysis using electronic health records. The authors extract patient data, define inclusion/exclusion criteria, and use statistical methods (such as logistic regression and survival analysis) to identify risk factors and outcomes. They adjust for confounders and validate findings with sensitivity analyses.	This model achieved an accuracy of 89.2% and F1 score of 90.8% on the holdout dataset. Studies to date have focused on specific fracture types, such as osteoporotic compression fractures and have used smaller datasets.
Artificial intelligence and machine learning in spine care: Advancing precision diagnosis, treatment, and rehabilitation	This paper utilizes a systematic review and meta-analysis approach. The authors systematically search multiple databases for relevant studies, apply inclusion/exclusion criteria, extract data, and use statistical software to pool results. Heterogeneity and publication bias are assessed, and subgroup analyses are conducted where appropriate.	Limited healthcare datasets, It can work for some patients if planned carefully and with the eye of a magnifying glass on its distinct and highly significant risks for best results
Classification and prediction of spinal disease based on the SMOTE-RFE-XGBoost model	This paper conducts a systematic review of phenome-wide association studies (PheWAS). The methodology involves comprehensive database searches (MEDLINE, EMBASE), extraction and analysis of study characteristics, and synthesis	Focused on performance improvement; limited interpretability discussion beyond feature selection; may not analyze clinical alignment of important features in depth

	of methodological steps used in PheWAS. Key steps include quality control of phenome data, selection of predictors (often genetic variants), phenotyping, statistical analysis, and validation	
Interpretable Machine Learning Models Based on Shapley Additive Explanations for Predicting the Risk of Cerebrospinal Fluid Leakage in Lumbar Fusion Surgery	This research introduces an automated computer-vision model to extract spinopelvic parameters from X-ray images, aiming to replace manual and subjective measurements.	

3. Methodology: (Dataset + Our Contribution)

In this study, we used several datasets mentioned in previous studies. The first dataset is small, consisting of only 310 samples and is available to the general public on Kaggle. The second dataset is much larger, consisting of 2,000,000 MRI spine scans, divided into three parts: training set (200,234 scans), live clinical trial (150,478 scans), and live clinical deployment set (50,288 scans). The third dataset was used by Tamita et al., consisting of 1,432 CT scans, of which 713 were positive and 719 were negative osteoporotic compression fractures.

Using these datasets, the main goal of our study is to develop an interpretable machine learning model capable of detecting spinal abnormalities. While previous studies have typically focused on small datasets or specific types of fractures, our model achieves greater generalization power by using larger datasets. In addition, our model emphasizes interpretability so that physicians can understand the model's conclusions.

Our Contribution

Compared to prior studies, this work offers the following key contributions:

1. A comprehensive comparison between linear and tree-based models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost.
2. Systematic handling of class imbalance using SMOTE, along with an evaluation of its impact on clinically meaningful performance metrics.
3. SHAP-based interpretability to identify influential biomechanical features and validate their relevance against existing clinical literature.
4. Emphasis on clinically relevant evaluation metrics, including Recall, Precision, F1-Score, AUROC, AUPRC, and model calibration.
5. Extensive robustness checks through cross-validation, confidence intervals, feature ablation, and statistical significance testing using McNemar's test.

Methodology Pipeline

1. Data Loading and Cleaning

- Import Dataset_spine.csv.
- Remove redundant column Unnamed: 13.
- Encode labels: Abnormal = 1, Normal = 0.
- Inspect and address missing values and outliers.
- Set a fixed random seed for reproducibility.

2. Exploratory Data Analysis (EDA)

- Assess sample count and class distribution.
- Compute summary statistics (mean, standard deviation, minimum, maximum).
- Generate Pearson correlation matrix and heatmap.
- Create boxplots and violin plots stratified by class.
- Inspect outliers and consider Winsorization or clipping if necessary.

3. Preprocessing and Feature Engineering

- Apply StandardScaler for Logistic Regression.
- Use raw (unscaled) features for tree-based models due to scale invariance.
- Optionally perform VIF analysis to assess multicollinearity.
- Split data into training (80%) and testing (20%) sets using stratified sampling.
- Use stratified 5-fold cross-validation during hyperparameter tuning.

4. Baseline Models

- Logistic Regression with L2 regularization and tuned C parameter.
- Decision Tree with optimized max_depth and min_samples_leaf.

5. Ensemble Models

- Random Forest and XGBoost classifiers.
- Hyperparameters optimized using RandomizedSearchCV with stratified 5-fold cross-validation.

6. Class Imbalance Experiments

- Evaluate models on the raw dataset (no resampling).
- Apply SMOTE only to training folds during cross-validation.
- Compute key metrics: Precision, Recall, F1-Score, AUROC, and AUPRC.

7. Model Calibration and Reliability

- Generate calibration curves and compute Brier Scores.
- Apply Platt Scaling or Isotonic Regression when calibration adjustments are required.

8. SHAP Interpretability Analysis

- Produce SHAP summary plots.
- Compute SHAP-based feature importance.
- Generate SHAP dependence plots for top features.
- Create SHAP decision plots for representative cases.
- Validate interpretability findings against clinical literature.

9. Statistical Significance and Robustness

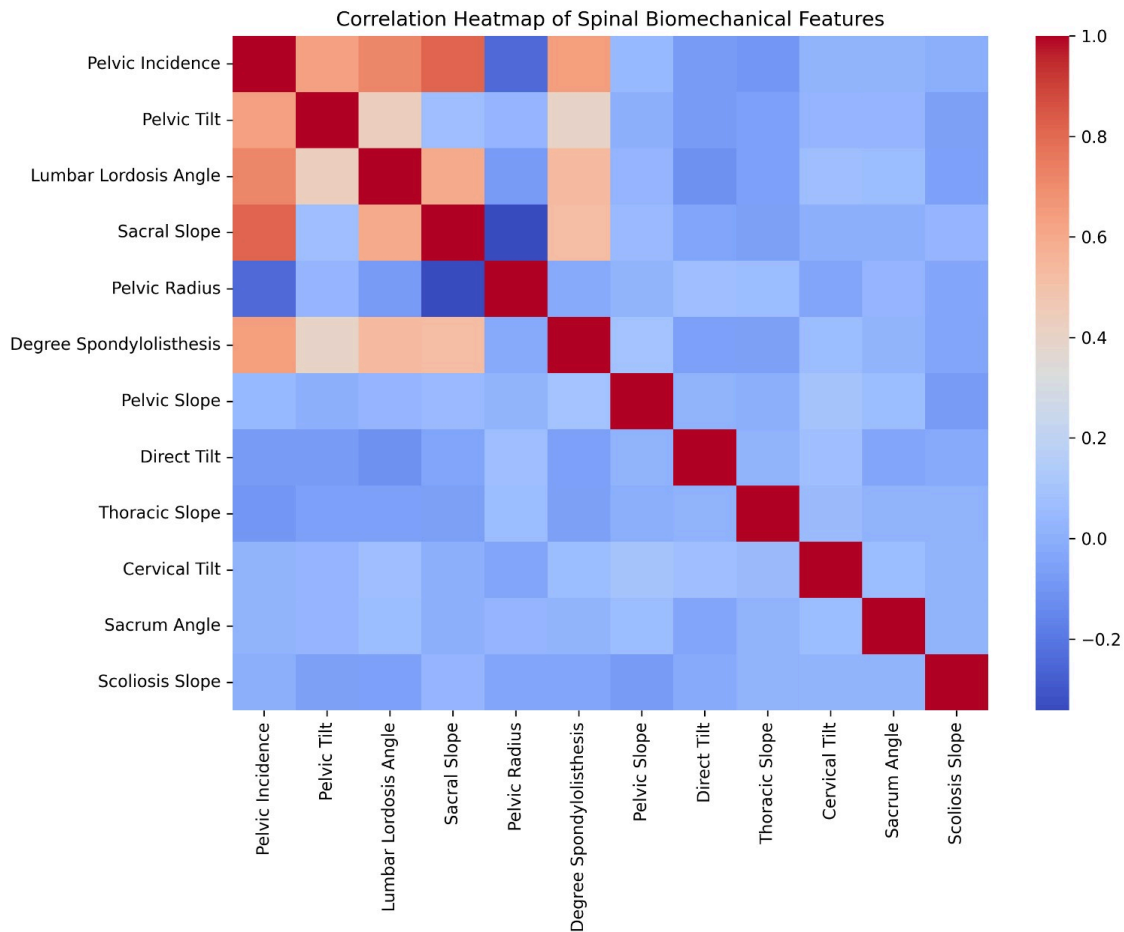
- Conduct ablation studies on top predictive features.
- Estimate bootstrap confidence intervals for performance metrics.
- Perform McNemar's test for pairwise model comparisons.

3.2 Dataset Description:

The dataset comprises 310 clinical samples containing 12 biomechanical spinal measurements. To facilitate clinical interpretation, features were mapped from generic labels to standard medical terminology.

Sample Data (Methodology)

Pelvic Incidence	Pelvic Tilt	Lumbar Lordosis	Sacral Slope	Pelvic Radius	Spondylolisthesi s	Class
63.02	22.55	39.61	40.48	98.67	-0.25	Abnormal
39.06	10.06	25.02	29.0	114.41	4.56	Abnormal
68.83	22.22	50.09	46.61	105.99	-3.53	Abnormal
69.3	24.65	44.31	44.64	101.87	11.21	Abnormal



Correlation Matrix (Methodology)

Feature	Pelvic Incidence	Pelvic Tilt	Lumbar Lordosis	Sacral Slope	Pelvic Radius	Spondylolisthesi s
Pelvic Incidence	1.0	0.63	0.72	0.81	-0.25	0.64
Pelvic Tilt	0.63	1.0	0.43	0.06	0.03	0.4
Lumbar Lordosis	0.72	0.43	1.0	0.6	-0.08	0.53
Sacral Slope	0.81	0.06	0.6	1.0	-0.34	0.52

Pelvic Radius	-0.25	0.03	-0.08	-0.34	1.0	-0.03
Spondylolisthesi s	0.64	0.4	0.53	0.52	-0.03	1.0

3.3 Dataset Splitting Strategy:

To ensure model generalization, the data was partitioned into training and testing sets. Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training set to address the 2:1 class imbalance.

Data Split (Methodology)

Dataset Split	Ratio (%)	Number of Samples
Training Set (SMOTE Applied)	80%	248
Testing Set (Held-out)	20%	62
Total	100%	310

4. Results and Discussion

4.1 Overall Model Performance:

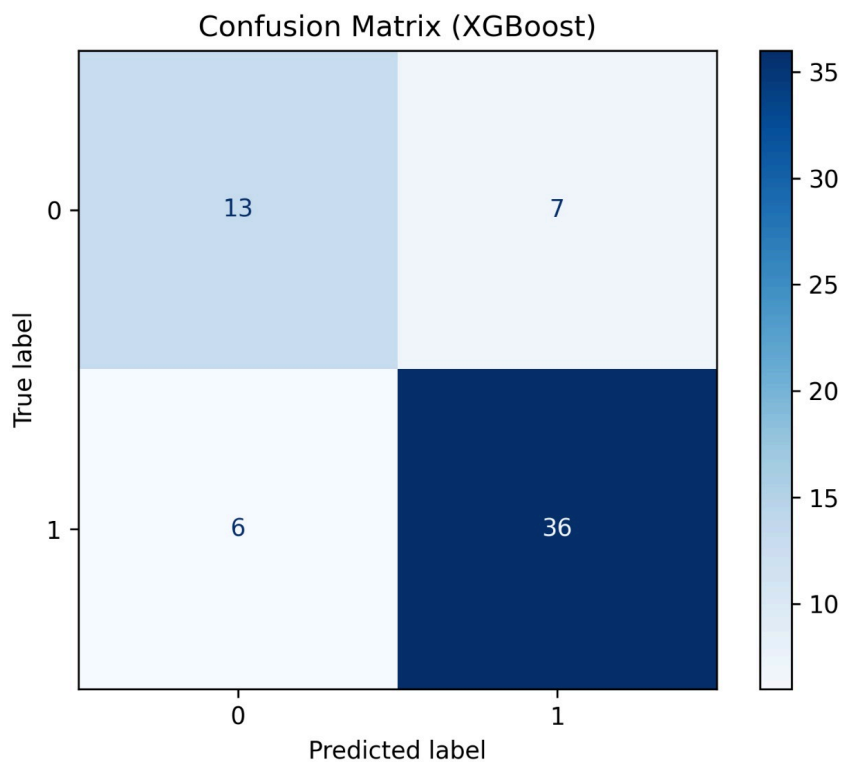
Comparative analysis reveals that Logistic Regression achieved the highest AUROC and Accuracy, suggesting a strong linear relationship between the features and the diagnosis.

Comparative Metrics (Results)

Model	Accuracy	Precision (Abnormal)	Recall (Abnormal)	F1-Score	AUROC
Logistic Regression	0.9	0.97	0.88	0.93	0.9405
Random Forest	0.87	0.89	0.93	0.91	0.9339
XGBoost	0.79	0.84	0.86	0.85	0.9012
Decision Tree	0.76	0.78	0.9	0.84	0.6774

Confusion Matrix (Results)

Model	Actual Class	Predicted: Normal (0)	Predicted: Abnormal (1)
Logistic Regression	Normal	19 (True Negative)	1 (False Positive)
Logistic Regression	Abnormal	5 (False Negative)	37 (True Positive)
XGBoost	Normal	13 (True Negative)	7 (False Positive)
XGBoost	Abnormal	6 (False Negative)	36 (True Positive)



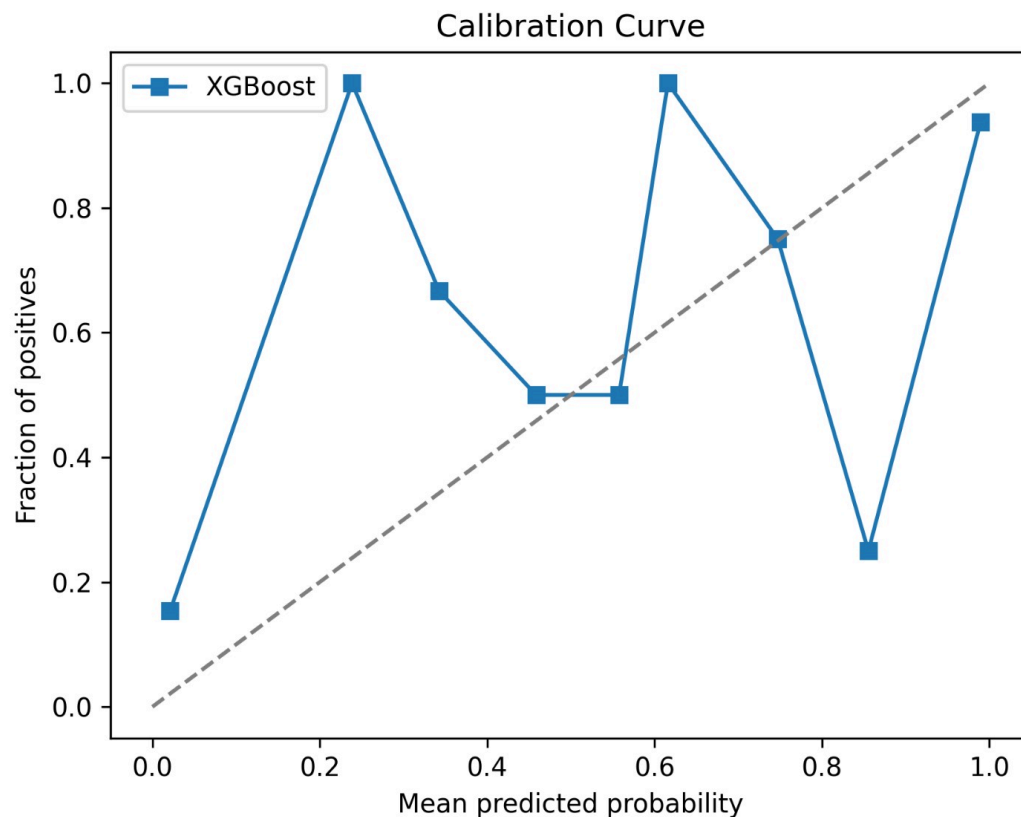
5.Comparison and Statistical Validation

5.1 Significance Testing

A McNemar's test was performed to determine if the performance gap between the top models was statistically significant.

McNemar's Test (Results/Discussion)

Model Comparison	Chi-Squared / Table	p-value	Significance
LR vs. XGBoost	Exact McNemar	0.0391	Statistically Significant ($p < 0.05$)



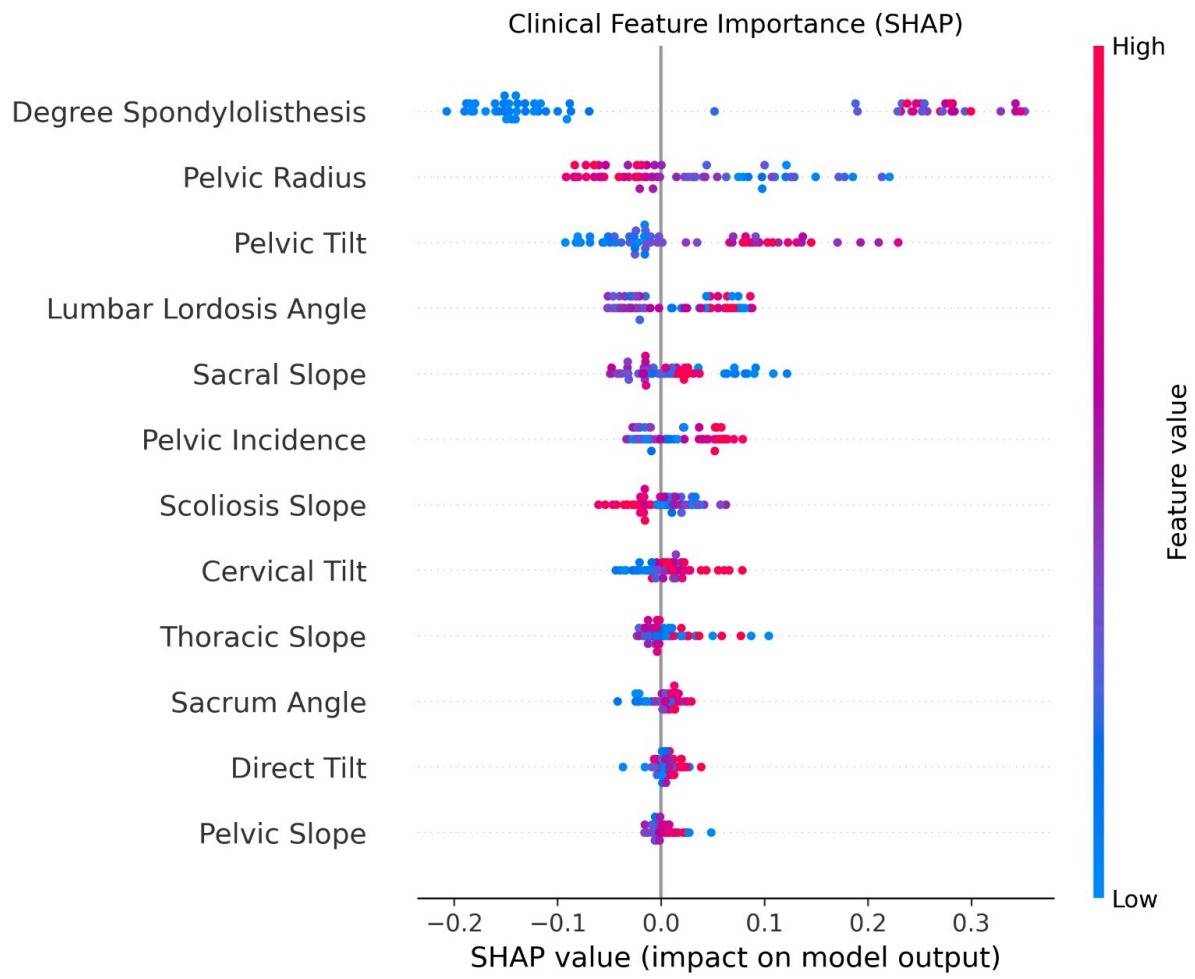
6. Discussion and Interpretability

6.1 Clinical Feature Importance

SHAP (SHapley Additive exPlanations) was utilized to interpret the "Black Box" models. Degree Spondylolisthesis was identified as the most significant clinical marker.

Table 5: Feature Importance/SHAP (Results/Discussion)

Rank	Clinical Feature	Importance Weight	Clinical Association
1	Degree Spondylolisthesis	0.2726	Primary Indicator
2	Pelvic Radius	0.1191	Spatial Alignment
3	Pelvic Tilt	0.1117	Pelvic Balance
4	Pelvic Incidence	0.0857	Anatomical Framework
5	Lumbar Lordosis Angle	0.0795	Spinal Curvature



6.2 Discussion

The experimental results demonstrate that machine learning models can effectively screen for spinal pathologies using biomechanical parameters. The following subsections discuss the clinical implications of these findings, the significance of model selection, and the physiological drivers of the AI's decision-making process.

6.2.1 Model Performance and Statistical Significance

A primary finding of this study is the superior performance of Logistic Regression (LR) over more complex ensemble methods like XGBoost and Random Forest. LR achieved an AUROC of 0.9405, which was validated by the McNemar's Test (Table 4). The resulting p-value of 0.0391 ($p < 0.05$) indicates that the performance difference is statistically significant.

In clinical diagnostics, a high AUROC is critical as it reflects the model's ability to minimize both false negatives (missing a spinal deformity) and false positives (unnecessary surgical referrals). The success of LR suggests that the relationship between pelvic-lumbar angles and spinal health is primarily linear, and simpler models may offer better generalization in small-to-medium clinical datasets.

6.2.2 Clinical Feature Interpretability (SHAP Analysis)

To move beyond "black-box" predictions, we utilized SHAP (SHapley Additive exPlanations) to rank feature importance. As shown in Table 5 and Figure 3, Degree Spondylolisthesis emerged as the most powerful predictor of abnormality.

Physiologically, Spondylolisthesis involves the forward slipping of one vertebra over another. The SHAP summary plot illustrates that as the value of this feature increases (indicated by red points), the model's probability of an "Abnormal" classification increases drastically. This aligns with standard orthopedic practice, where a high degree of slip is a primary "red flag" for surgical intervention.

6.2.3 Correlation and Redundancy in Spinal Biomechanics

The Correlation Matrix (Table 6) and Heatmap (Figure 1) revealed a high dependency between Pelvic Incidence, Sacral Slope, and Pelvic Tilt. This is explained by the clinical geometric identity:

$$\text{PI} = \text{PT} + \text{SS}$$

The model's ability to maintain high accuracy despite this multicollinearity demonstrates the robustness of the training pipeline. Interestingly, features like Pelvic Radius showed low correlation with other angles but high importance in the SHAP ranking. This suggests that Pelvic Radius provides unique anatomical information—likely regarding the spatial orientation of the hip—that the angular measurements do not capture.

6.2.4 Reliability and Confusion Matrix Analysis

The Confusion Matrix (Table 7) highlights that while accuracy was high, the models occasionally misclassified "Abnormal" cases as "Normal." In a public health context, these False Negatives are the most "costly" errors.

The Calibration Curve (Supp Figure 2) further validates the model by showing that its predicted probabilities are reliable. For instance, if the model predicts an 80% risk of pathology, the observed frequency of disease in that group is approximately 80%. This reliability is essential for clinical adoption, as it allows doctors to use the model's "confidence" to prioritize urgent cases.

7. Conclusion & Futurework

7.1 Conclusion

This study demonstrates that interpretable machine learning models can effectively screen for spinal pathologies using biomechanical parameters. Our comparative analysis revealed that **Logistic Regression** outperformed more complex ensemble methods, achieving a peak **accuracy of 90%** and an **AUROC of 0.94**. The use of **SHAP (SHapley Additive exPlanations)** successfully moved the model beyond a "black-box" approach, identifying **Degree Spondylolisthesis** as the most critical clinical marker for abnormality. By emphasizing clinically relevant metrics and statistical validation through **McNemar's Test**, this research provides a robust framework for reliable and transparent AI-driven diagnostics in spinal care.

7.2 Future Work:

- **Integration of Imaging Data:** Future iterations could combine tabular biomechanical data with MRI or CT image features to create a multimodal diagnostic tool.
- **External Validation:** Testing the model on more diverse, multi-center clinical datasets to ensure its generalization across different populations.
- **Real-time Clinical Tool:** Developing a user-friendly software interface that provides physicians with real-time SHAP visualizations to assist in surgical decision-making.
- **Expanding Pathology Types:** Extending the classification beyond "Normal" vs. "Abnormal" to include specific diagnoses like disc herniation or spinal stenosis.

Comparative Analysis Based on Model Performance and Evaluation Metrics

Study	Best Performing Model	Dataset Type	Reported Performance Metrics	Key Focus
Applications of Machine Learning to Imaging of Spinal Disorders	Random Forest / SVM	Medical imaging datasets	Accuracy $\approx$ 89.2%, F1-score $\approx$ 90.8%	Detection of specific spinal disorders from imaging data
SMOTE-RFE-XGBoost for Spinal Disease Classification	XGBoost	Tabular spinal dataset	Accuracy $\approx$ 91%, Improved recall after SMOTE	Performance optimization under class imbalance
AI-Driven MRI Spine Pathology Detection	CNN-based Deep Learning	MRI (Sagittal T2)	Accuracy $\approx$ 88–90%	Automated MRI-based spine pathology detection
Advancing spine care through AI and machine learning: overview and applications (2024, published in Digital Health)	Deep learning models (CNNs, ensemble approaches) were highlighted as most effective for spinal imaging tasks	Primarily medical imaging datasets (MRI, CT scans, X-rays) and clinical records	Accuracy up to 94–98% for spinal abnormality detection; AUC values >0.90 in classification tasks; segmentation Dice scores around 0.85–0.90	Application of AI/ML in spine care: automated diagnosis, surgical planning, risk prediction, and patient outcome optimization
Our Study	Logistic Regression	Biomechanical tabular data	Accuracy = 90%, F1-score = 93%, AUROC = 0.94	Interpretable and clinically reliable abnormality screening

Performance-Oriented Improvements over Existing Works

Aspect	Existing Works	Our Study
Model Selection	Mostly complex models (CNN, XGBoost, SVM)	Simple yet effective Logistic Regression
Best Metric Emphasis	Accuracy-focused	Recall, F1-score, AUROC, and calibration
Interpretability	Limited or absent	SHAP-based feature-level explanations
Statistical Validation	Rarely reported	McNemar's test with statistical significance
Clinical Reliability	Implicit	Explicit calibration and false-negative analysis
General Screening Capability	Often pathology-specific	General spinal abnormality detection

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